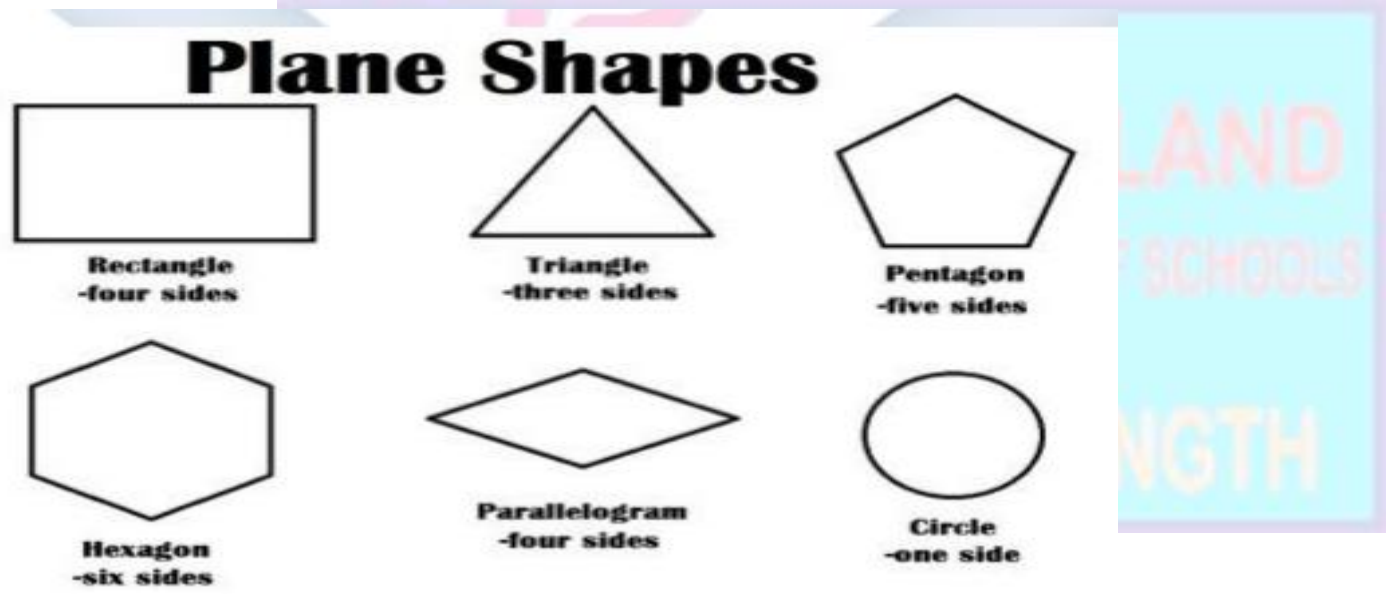
JSS 1 MATHEMATICS LESSON NOTES (THIRD TERM)

WEEK 2: PLANE SHAPES

What Are Plane Shapes?

Plane shapes (or **2D shapes**) are flat shapes that have only two dimensions: **length and width**. They have **no thickness**.



Examples of Common Plane Shapes

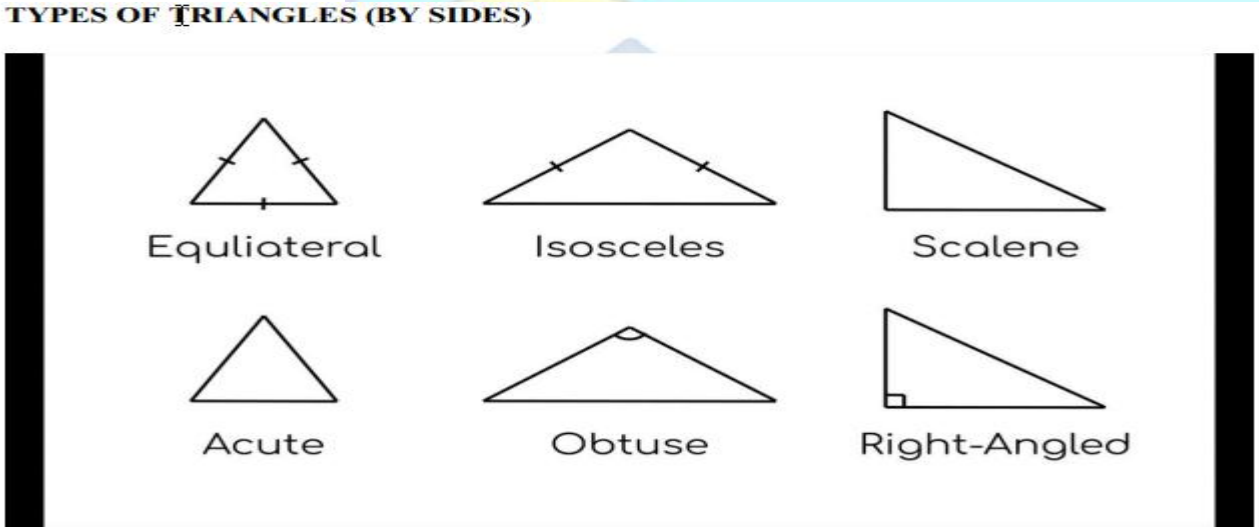
- Triangles
- Quadrilaterals (square, rectangle, parallelogram, rhombus, trapezium)

- Circles
- Polygons

TRIANGLES

A triangle is a plane shape with **three sides and three angles**.

Types of Triangles (By Sides)



1. Equilateral Triangle

- All three sides are equal
- All three angles are equal (60° each)
- Has 3 lines of symmetry

Properties:

- $AB = BC = AC$
- $\angle A = \angle B = \angle C = 60°$

2. Isosceles Triangle

- Two sides are equal
- Two angles are equal (opposite equal sides)
- Has 1 line of symmetry

Property:

- If $AB = AC$, then $\angle B = \angle C$

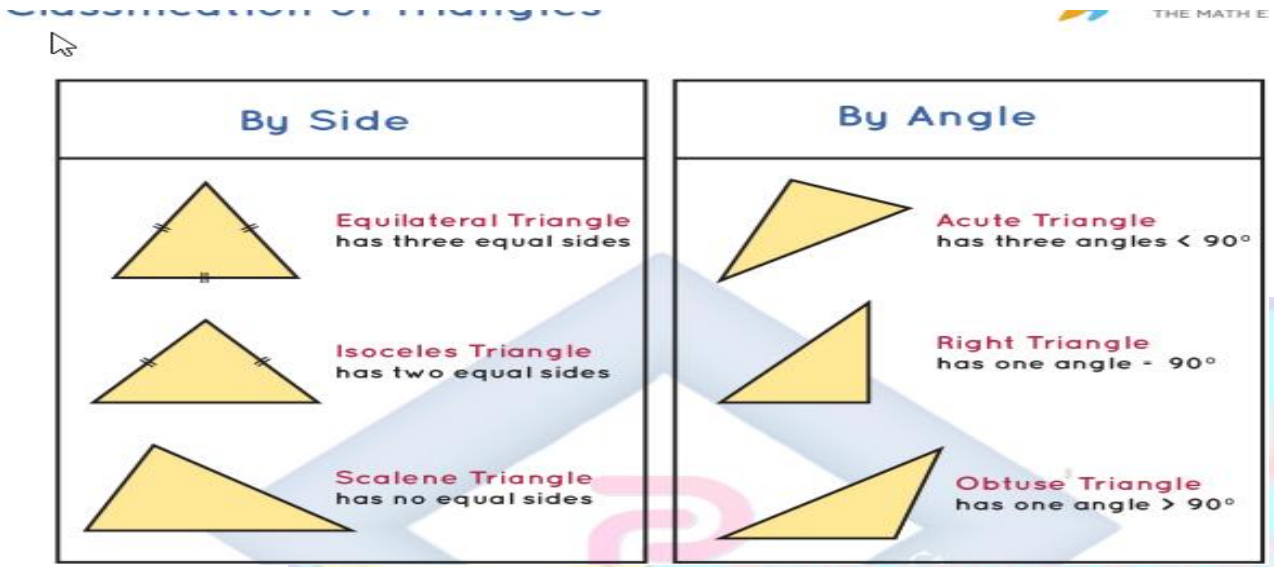
3. Scalene Triangle

- All sides are different
- All angles are different
- No line of symmetry

Properties:

- $AB \neq BC \neq AC$
- $\angle A \neq \angle B \neq \angle C$

Types of Triangles (By Angles)



1. Acute-Angled Triangle

- All angles are less than 90°

2. Right-Angled Triangle

- One angle is 90°
- Other two angles are acute
- Hypotenuse is opposite the right angle

3. Obtuse-Angled Triangle

- One angle is greater than 90°

- Other two angles are acute

Properties of Triangles

1. **Sum of angles:**
   $\angle A + \angle B + \angle C = 180°$
2. **Sum of two sides:**
   Sum of any two sides is greater than the third side
3. **Perimeter:**
   Perimeter = a + b + c
4. **Area:**
   Area = ½ × base × height

Examples

Example 1:
Find the third angle if two angles are 50° and 70°
Solution:
50° + 70° = 120°
Third angle = 180° − 120° = **60°**

Example 2:
Sides = 5 cm, 7 cm, 9 cm
Perimeter = 5 + 7 + 9 = **21 cm**

Example 3:
Base = 8 cm, Height = 6 cm
Area = ½ × 8 × 6 = **24 cm²**

QUADRILATERALS

A quadrilateral is a shape with **four sides and four angles**.

Types of Quadrilaterals

1. Square

Properties:

- All sides equal
- All angles are 90°
- Opposite sides are parallel
- Diagonals bisect at right angles
- 4 lines of symmetry

Formulas:

- Perimeter = 4s
- Area = s^2

Example:
Side = 7 cm
Perimeter = 28 cm
Area = 49 cm^2

2. Rectangle

Properties:

- Opposite sides equal and parallel
- All angles are 90°
- Diagonals are equal

Formulas:

- Perimeter = 2(l + w)
- Area = l × w

Example:
l = 12 cm, w = 5 cm
Perimeter = 34 cm
Area = 60 cm²

3. Parallelogram

Properties:

- Opposite sides equal and parallel
- Opposite angles equal
- Diagonals bisect each other

Formulas:

- Perimeter = 2(a + b)
- Area = base × height

4. Rhombus

Properties:

- All sides equal
- Opposite angles equal
- Diagonals intersect at right angles

Formulas:

- Perimeter = 4s
- Area = ½ × d_1 × d_2

5. Trapezium

Properties:

- One pair of sides parallel
- Parallel sides are called bases

Formula:

- Area = ½ × (a + b) × h

6. Kite

Properties:

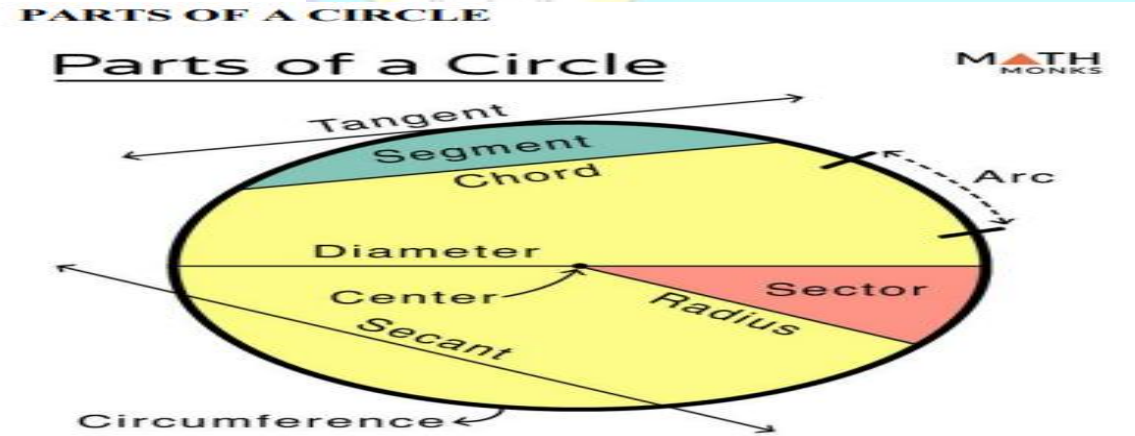
- Two pairs of adjacent equal sides
- One pair of opposite angles equal
- Diagonals meet at right angles

Summary Table

Quadrilateral	Equal Sides	Right Angles	Parallel Sides
Square	All 4	All 4	Both pairs
Rectangle	Opposite	All 4	Both pairs
Parallelogram	Opposite	None	Both pairs
Rhombus	All 4	None	Both pairs
Trapezium	None	None	One pair
Kite	Adjacent pairs	None	None

CIRCLES

A circle is a shape where all points are the same distance from a fixed point (**center**).



Parts of a Circle

- Center
- Radius (r)
- Diameter ($d = 2r$)
- Chord
- Arc
- Circumference
- Sector
- Segment
- Tangent

Circle Formulas

- Circumference:
  $C = 2\pi r$ or $C = \pi d$
- Area:
  $A = \pi r^2$

($\pi \approx 3.14$ or $22/7$)

Examples

Example 1:
$r = 7$ cm
$C = 44$ cm

Example 2:
r = 14 cm
A = 616 cm^2

Example 3:
d = 20 cm
C = 62.8 cm

Example 4:
C = 44 cm
r = 7 cm

Relationship Between Shapes

- A square is a special rectangle
- A square is a special rhombus
- A rectangle is a parallelogram
- A rhombus is a parallelogram

1. **Classwork**
2. What is the sum of angles in a triangle?
3. Find the perimeter of a rectangle (15 cm, 8 cm)
4. Find area of a square (side = 9 cm)
5. Find circumference of a circle (r = 21 cm, π = 22/7)
6. State two properties of a rhombus

Assignment

1. Two angles are 45° and 60°. Find the third

2. Rectangle: l = 20 cm, w = 12 cm → Find area
3. Square area = 64 cm² → Find perimeter
4. Circle diameter = 14 cm → Find area
5. Trapezium: a = 10 cm, b = 16 cm, h = 8 cm → Find area

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GOD IS OUR STRENGTH